Note: Actually, the Exam Preparation was modified from the past year exercise called "Group Works", thanks to the past semester CEM's TA as mentioned below.

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1. Find a linear combination of the vector $\mathbf{v} = (1, 2, 3)$ and $\mathbf{w} = (-2, 3, -1)$ in $\mathbb{R}^3$ that adds up to (8, -5, 9)

2. Show that (1, 0, 0) and (16, 25, 10) are both eigenvectors of the matrix $\mathbf{B}$ (mentioned below).
What are the corresponding eigenvalues? Is (0, 2, 3) an eigenvector? How about (0, -3, 2)?

$$\mathbf{B} = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix}$$

3. Suppose $\mathbf{x} \in \mathbf{R}^m$ is an eigenvector of an $m \times m$ matrix $\mathbf{A}$.
Show that if $k \in \mathbf{R}$ is any scalar, then $k\mathbf{x}$ is also an eigenvector of $\mathbf{A}$, and has the same eigenvalue as $\mathbf{x}$.

4. Define a function $f\colon X \Rightarrow Y$ whose domain and range both contain just the first five letter of the alphabets: $X = Y = \{a, b, c, d, e\}$. Define the "rule" f for this function by setting:

$$f(a) = b,\ f(b) = c,\ f(c) = a,\ f(d) = b,\ f(e) = c$$

   4.1. Find the images of these sets: $\{a, b, c\}$, $\{a, b, d\}$, $\{a, b, e\}$
   4.2. Find the pre-images of these sets: $\{a\}$, $\{a, b\}$, $\{a, b, c\}$, $\{c\}$, and $\{d, e\}$.
   4.3. What is the image of f?
   4.4. Is this a function one-to-one? Is it onto? Explain
   4.5. Define a function $g\colon X \Rightarrow Y$ (same X and Y as above) which is one-to-one, onto, and satisfies $g(a) = c$.

5. Consider the mapping $F\colon \mathbf{R}^3 \Rightarrow \mathbf{R}^2$ given by $S(x, y, z) = (z, x)$.
   5.1. Which points in $\mathbf{R}^2$ have non-empty pre-images?
   5.2. How would you describe the pre-image of a typical point in the range?
   5.3. Is F onto? Is F one-to-one?

6. Suppose k is a scalar, and $T_k\colon \mathbf{R}^m \Rightarrow \mathbf{R}^m$ is scalar multiplication by k. Thus, $T_x(\mathbf{x}) = k\mathbf{x}$ for any $\mathbf{x} \in \mathbf{R}^m$. Show that T_k is linear.

7. Show that a function $f\colon \mathbf{R} \Rightarrow \mathbf{R}$ of the form $f(x) = mx + b$, though often called "linear" in other contexts, does not meet the linearity criteria of Definition 4.1 (and hence is not linear in our terminology) unless $b = 0$.

8. Consider the transformations $\mathbf{R}^3 \Rightarrow \mathbf{R}^2$ given by
   a) $T(x, y, z) = (|x|, |z|)$
   b) $P(x, y, z) = (x, z)$
   Prove that P is linear, but T is not.

9. What matrix represents the projection $P\colon \mathbf{R}^4 \Rightarrow \mathbf{R}^2$ given by $P(x_1, x_2, x_3, x_4) = (x_2, x_3)$?

10. What matrix represents the injection $Q\colon \mathbf{R}^2 \Rightarrow \mathbf{R}^4$ given by $P(x, y) = (0, x, y, 0)$?